📜 MASTER AI GLOSSARY — COMPLETE CONTENT---------Raw Doc by Jolly Gupta

Total: 502 Terms · 30 Categories

CORE AI (22 terms)

Term	Description
Affective Computing	AI recognizing, interpreting, and simulating human emotions. Used in healthcare, customer service, HCI.
Algorithm	Finite step-by-step procedure for solving a problem. All AI systems are built on algorithms.
Ambient Intelligence	Environments embedded with responsive AI — smart homes, assisted living, adaptive public spaces.
Artificial Intelligence (AI)	Machines performing tasks requiring human-like intelligence — reasoning, learning, language, decisions.
Augmented Reality (AR)	Overlaying AI-generated information onto the real world across visual and other sensory modalities.
Chatbot	Software simulating conversation via text or voice. Modern chatbots run on large language models.
Cognitive Computing	Systems mimicking human thought — perception, reasoning, learning — to support human decisions.
Composite AI	Combining ML + NLP + rules + knowledge graphs to solve complex problems no single approach handles.
Context Length	Actual number of tokens in current conversation — different from maximum context window size.
Context Window	Maximum text AI processes at once — its working memory. Larger = handles longer documents.
Expert System	AI emulating domain expert decisions using if-then rules. Still used in some regulated industries.
Foundation Model	Large general-purpose AI trained on massive data, adaptable for many tasks. Claude, GPT-4, Gemini.
Frontier Model	Most capable AI models at any given time — cutting edge of what is technically possible.
Large Language Model (LLM)	Massive neural network trained on internet-scale text. Understands and generates human language.

Machine Perception	Computer system interpreting sensory data — vision, audio, touch — analogous to human perception.
Model	Trained AI system storing learned patterns as numerical parameters. Output of training; input to inference.
Multimodal AI	AI processing and generating text, images, audio, video in one unified system.
Multimodality	Capacity to understand and generate across multiple data types simultaneously.
Narrow AI	AI for one specific task only. All current AI including advanced LLMs is narrow AI.
Parameter	Learned numbers storing encoded knowledge. More parameters = generally larger, more capable model.
Reasoning	AI logically working through problems step by step rather than just pattern-matching.
Small Language Model (SLM)	Compact LLMs for on-device deployment. Examples: Phi-3, Gemma 2B, Mistral 7B.

CONCEPT (19 terms)

Term	Description
AGI (Artificial General Intelligence)	Hypothetical AI matching human cognitive ability across all domains. Doesn't exist yet.
Augmentation	Using AI to enhance human capability rather than replace humans.
Automation	Tasks performed by AI with little or no human involvement.
Black Box	AI model whose internal reasoning is opaque — we see inputs and outputs but not the process.
Closed Source Model	AI with private weights, accessible only via API. GPT-4, Claude, Gemini.
Context Rot	Performance degradation in long conversations as AI loses track of earlier context and instructions.
Emergent Behavior	Unexpected capabilities appearing in large AI as models scale — not explicitly trained for.
General AI	Hypothetical AI with human-level intelligence across diverse domains. Does not exist yet.
Intelligence Amplification	Using technology to augment human intelligence rather than replace it.

Intelligence Explosion	Hypothetical scenario where AI recursively self-improves leading to rapid superintelligence emergence.
Knowledge Cutoff	Date after which AI has no information. Training data ends there.
Lost in the Middle	LLM failure where information in the middle of long contexts gets less attention than start or end.
Open Source Model	AI with publicly available weights. LLaMA, Mistral, Falcon. Enables local deployment.
Quantum Computing	Computing using superposition and entanglement. Potential to accelerate certain AI/ML tasks dramatically.
Superintelligent AI	Hypothetical AI vastly surpassing human intelligence. Central concern of AI safety research.
Technological Singularity	Hypothetical future where AI growth becomes uncontrollable, leading to profound civilization change.
Transhumanism	Philosophical movement advocating technology to enhance human capabilities beyond biological limits.
Turing Test	Alan Turing's 1950 test — if humans can't distinguish AI from human in conversation, AI passes.
Weak AI	Another term for narrow AI — focused on one task only. All currently deployed AI.

MACHINE LEARNING (42 terms)

Term	Description
Adaptive Algorithm	Changes behavior at runtime based on a reward or feedback signal.
Bagging / Bootstrap Aggregating	Training multiple models on random data subsets, averaging predictions. Reduces variance.
Bias-Variance Tradeoff	Balance between underfitting (high bias) and overfitting (high variance).
Boosting	Sequential ensemble where each model corrects errors of previous. XGBoost uses this.
Classification	Assigning inputs to predefined categories — spam detection, disease diagnosis.
Cluster Analysis	Grouping similar data without labels. Used for segmentation and anomaly detection.
Decision Tree	Flowchart-like model making decisions at feature-value nodes. Interpretable but weak alone.

Dimensionality Reduction	Reducing input variables while preserving information. PCA, t-SNE, UMAP.
Discriminative Model	Learns decision boundaries between classes. Logistic regression, SVM.
Double Descent	Very large models outperform medium ones — contradicts classical overfitting theory.
Ensemble Learning	Combining multiple models for better predictions than any single model.
Generalization	Model performing well on unseen data beyond training examples. The ML objective.
Gradient Boosting	Sequential ensemble correcting residuals. XGBoost, LightGBM, CatBoost.
Incremental Learning	Continuously updates model with new data without full retraining.
K-Means Clustering	Partitions data into K clusters by minimizing distance to centroids.
K-Nearest Neighbors (KNN)	Classifies by finding K most similar training examples. No training phase.
Kernel Method	Class of ML algorithms using kernel functions for pattern analysis. SVM is the key example.
Lazy Learning	Delays generalization until query time. KNN is the canonical example.
Machine Learning (ML)	Computers learning patterns from data without explicit programming.
Markov Decision Process (MDP)	Mathematical framework for sequential decision-making under uncertainty. Foundation of RL.
Naive Bayes Classifier	Probabilistic classifier applying Bayes theorem with independence assumptions between features.
Offline Learning	Training on a fixed dataset — the traditional batch training approach.
Online Machine Learning	Model updates continuously as new data arrives in real time.
Overfitting	Model memorizes training data — great on training, poor on new data.
Pattern Recognition	Automatic discovery of regularities in data to classify or predict.
Predictive Analytics	Using historical data to predict future or unknown events. Core of business AI.
Principal Component Analysis (PCA)	Reduces dimensionality by finding axes of maximum variance.
Proximal Policy Optimization (PPO)	Popular RL algorithm. Widely used in RLHF pipelines for LLM fine-tuning.

Q-Learning	Model-free RL learning optimal action values per state. Foundation of deep RL.
Random Forest	Ensemble of decision trees. Robust, interpretable, handles missing data.
Regression Analysis	Estimating relationships between variables to predict continuous values.
Reinforcement Learning	Agent learns via trial-and-error with reward signals. Powers game AI, robotics, RLHF.
Representation Learning	Learning useful feature representations from raw data — deep learning automates this.
SARSA	On-policy RL updating values based on actual next action taken. Contrast with Q-learning.
Self-Supervised Learning	Uses data's own structure to generate labels. LLM next-token prediction is self-supervised.
Semi-Supervised Learning	Mix of labeled and unlabeled data. Useful when labeling is expensive.
Similarity Learning	Learning a similarity function between objects. Used in recommendation, face verification.
Supervised Learning	Training on labeled input-output pairs. Most common ML paradigm.
Support Vector Machine (SVM)	Finds optimal decision hyperplane. Powerful for small high-dimensional datasets.
Temporal Difference Learning	RL updating value estimates based on differences between successive predictions. Foundation of Q-learning.
Underfitting	Model too simple to capture patterns — poor even on training data.
Unsupervised Learning	Learning from unlabeled data to discover hidden patterns.

DEEP LEARNING (14 terms)

Term	Description
Batch Normalization	Normalizes layer inputs during training. Speeds convergence, stabilizes deep networks.
CNN (Convolutional Neural Network)	Uses filters to extract spatial features from images. Dominant for image tasks.
Capsule Neural Network (CapsNet)	Models hierarchical spatial relationships better than CNNs by grouping neurons into capsules.

Deep Learning	ML using multi-layered neural networks for unstructured data. Powers modern AI.
Dropout	Randomly disables neurons during training to prevent co-adaptation and overfitting.
Hidden Layer	Layers between input and output. More layers = more capacity for complex features.
LSTM (Long Short-Term Memory)	RNN variant with gating for long-term dependencies. Largely replaced by transformers.
Latent Space	Internal space where AI represents concepts as vectors. Semantic arithmetic works here.
Multilayer Perceptron (MLP)	Fully-connected feedforward neural network. The simplest deep learning architecture.
Neural Network	System of layered nodes inspired by the brain. Learns patterns through training.
Perceptron	Simplest neural network — a single linear classifier. Historical starting point of DL.
Recurrent Neural Network (RNN)	Feedback loops for sequential data. Largely replaced by transformers.
Reservoir Computing	Fixed random recurrent network maps inputs to higher dimensions — only output layer trains.
Spiking Neural Network (SNN)	Mimics biological neurons with timing and spike-based communication between nodes.

ARCHITECTURE (22 terms)

Term	Description
Activation Function	Decides whether a neuron fires. Introduces non-linearity. ReLU, sigmoid, GELU.
Attention Mechanism	Dynamically weights relevant input parts when generating output. Core of transformers.
Autoencoder	Compresses input to smaller representation then reconstructs. Used for denoising and feature learning.
Autoregressive Model	Generates output one token at a time, each conditioned on previous tokens. GPT-style.
Backpropagation	Propagates errors backward to update weights via gradient descent. The fundamental learning algorithm.

Boltzmann Machine	Stochastic recurrent neural network learning probability distributions. Precursor to modern generative models.
Decoder Model	Generates text token by token using previous tokens. GPT family. The basis of chat AI.
Encoder Model	Converts input to contextual embeddings. BERT, RoBERTa. Used for understanding, not generation.
Encoder-Decoder Model	Encodes input then decodes to output. T5, BART. Ideal for translation and summarization.
Flash Attention	Memory-efficient, hardware-optimized attention algorithm. Faster, less GPU memory.
Mixture of Experts (MoE)	Routes inputs to specialized sub-networks. More efficient at scale. GPT-4, Mixtral.
Neural Turing Machine	RNN with external memory — can learn sorting and copying from examples.
Neuromorphic Computing	Hardware mimicking neural brain structures. Ultra-low-power AI at the edge.
Neuron	Basic processing unit receiving weighted inputs, applying activation, passing output forward.
Positional Encoding	Tells transformers the order of tokens — necessary because attention has no sequence awareness.
Restricted Boltzmann Machine (RBM)	Generative stochastic network learning probability distributions. Building block of early deep learning.
RoPE (Rotary Position Embedding)	Advanced positional encoding enabling context length extrapolation. LLaMA, Mistral, Qwen.
Seq2Seq	Encoder-decoder converting one sequence to another. Foundation of translation and summarization.
Softmax	Converts raw scores to probabilities summing to 1. Used in classification and attention.
Transformer	Attention-based architecture introduced by Google in 2017. Powers all modern LLMs.
Vision Transformer (ViT)	Transformer applied to images by treating patches as tokens. Rival to CNNs.
Weights	Adjustable parameters encoding learned knowledge. Updated by backpropagation.

NLP (17 terms)

Term	Description
Bag-of-Words Model	Represents text as unordered word frequency counts — ignoring grammar and word order.
Computational Linguistics	Applying computational methods to model human language. Foundation of modern NLP.
Dialogue System	Computer system designed to converse coherently with humans across multiple turns.
Embedding	Dense numerical vector representing meaning. Similar concepts cluster nearby in vector space.
Language Model	Probabilistic model assigning probabilities to text sequences. Foundation of all LLM systems.
Named Entity Recognition (NER)	Locates and classifies people, organizations, locations in unstructured text.
Natural Language Generation (NLG)	AI producing human-readable text — articles, summaries, responses.
Natural Language Processing (NLP)	Field enabling computers to understand, interpret, and generate human language.
Natural Language Understanding (NLU)	AI grasping meaning, intent, and context — not just surface-level word matching.
Neural Machine Translation	End-to-end neural translation between languages. Replaced older statistical methods.
Prompt	Input text given to an AI model. Quality of prompt is the biggest factor in output quality.
Sentiment Analysis	Classifying text emotion as positive, negative, or neutral.
Speech Recognition	AI converting spoken words to text. Powers Siri, Alexa, and medical dictation.
Summarization	Condensing long documents. Extractive (selects) vs. abstractive (generates new text).
Token	Basic unit LLMs process — roughly a word or subword. All costs and limits measured in tokens.
Tokenization	Splitting text into tokens before model processing. Affects multilingual ability.
Word Embedding	Dense vector representing a word's meaning as geometry. king - man + woman = queen.

GENAI (5 terms)

Term	Description
Deepfake	AI-generated synthetic media convincingly mimicking real people. Major fraud and misinformation risk.
Diffusion Model	Creates images by reversing a noise-adding process. Powers DALL-E, Stable Diffusion, Midjourney.
GAN (Generative Adversarial Network)	Two competing networks — generator creates, discriminator judges. Produces realistic synthetic data.
Generative AI	AI creating new content — text, images, video, music, code — from learned patterns.
Puppeteering	AI animating full-body images to perform actions — a form of body deepfake.

RAG (16 terms)

Term	Description
Agentic RAG	Agent dynamically decides what to retrieve, when, and how many times based on task needs.
BM25	Classic keyword-based document ranking algorithm. Fast, used in hybrid RAG alongside vector search.
Chunk Overlap	Adjacent chunks share some content to prevent losing context at boundaries.
Chunking	Splitting documents into smaller pieces before embedding. Chunk size impacts retrieval quality.
Context Stuffing	Packing as many relevant chunks as possible into the LLM context window.
Dense Retrieval	Retrieves by vector similarity. Understands meaning. More accurate but slower than BM25.
Embedding Model	Model converting text to vectors for search. OpenAI ada, Cohere embed, BAAI/bge.
GraphRAG	RAG using a knowledge graph instead of vector database. Better for multi-hop reasoning.
Grounding	Connecting AI outputs to verified external facts to reduce hallucinations.

Hybrid Search	Combining BM25 keyword search with vector search. Better recall than either alone.
RAG (Retrieval-Augmented Generation)	AI searches external knowledge before answering. Solves knowledge cutoff and reduces hallucinations.
Re-ranking	Second-stage scoring of retrieved documents before passing to LLM. Improves RAG precision.
Retrieval	Looking up relevant information from a knowledge base to ground AI responses.
Semantic Search	Search understanding query meaning via embeddings. Finds relevant results without exact keyword match.
Sparse Retrieval	Retrieves by keyword matching (BM25, TF-IDF). Fast, interpretable, misses semantic meaning.
Vector Database	Stores and searches embeddings efficiently. Core infrastructure for RAG systems.

PROMPTING (14 terms)

Term	Description
Chain-of-Thought (CoT)	Asks AI to reason step by step before answering. Dramatically improves complex task accuracy.
Delimiters	Special characters separating sections in a prompt. Triple backticks, XML tags.
Few-Shot Learning	Providing a few examples in the prompt to guide format or style without retraining.
In-Context Learning	AI learning from examples in the prompt without weight updates. Emergent capability of large models.
Meta-Prompt	Prompt that generates or optimizes other prompts. Used in automated prompt engineering.
One-Shot Learning	Giving AI exactly one example in the prompt to guide output format or style.
Prompt Engineering	Systematic design combining instructions, examples, formatting, personas to maximize output quality.
Role Prompting	Instructing AI to adopt a persona — shapes tone, expertise level, and response style.
Self-Consistency	Running same prompt multiple times and selecting most common answer. Reduces variance.

Structured Output	Forcing AI to respond in JSON, XML, or other formats. Essential for programmatic consumption.
System Prompt	Hidden instructions before conversation begins — sets persona, rules, tone, constraints.
Tree of Thought (ToT)	Model explores multiple reasoning branches and evaluates them before committing to an answer.
Zero-Shot Learning	AI completing tasks never explicitly trained on — generalizing from broad pretraining.
Zero-Shot Prompting	Asking AI to perform a task with only an instruction and no examples.

AGENTS (13 terms)

Term	Description
Agent / AI Agent	Application achieving a goal by processing input, reasoning with tools, and taking autonomous actions.
Agentic AI	AI systems that plan, decide, and execute multi-step tasks autonomously.
Behavior Tree	Hierarchical model organizing complex agent behaviors from simple modular tasks. Used in game AI and robotics.
Distributed AI	Multiple agents collaborating across networks without centralized control.
Function Calling	Structured way for LLMs to request specific function outputs from external systems with defined schemas.
Intelligent Personal Assistant	Software agent performing tasks based on voice or text commands. Siri, Alexa, Google Assistant.
Memory (AI Agents)	How agents store and recall information — short-term (context), long-term (vector DB), episodic (history).
Multi-Agent System	Multiple AI agents collaborating or competing to solve problems no single agent could handle.
Orchestrator	Controller deciding which agents to call, in what order, and how to combine outputs.
Planning (AI)	AI breaking a complex goal into ordered sub-tasks and sequencing them for execution.
ReAct (Reasoning + Acting)	Framework alternating reasoning steps and tool actions. Improves agent reliability.
Scratchpad	Internal space where AI writes intermediate reasoning before producing a final answer.

Tool Use	AI's ability to call external tools — calculators, APIs, search, code interpreters.

CACHING (6 terms)

Term	Description
Context Rot	Performance degradation in long conversations as AI loses track of earlier context.
Embedded Cache	Storing pre-computed embeddings so they don't need to be regenerated on every query.
KV Cache (Key-Value Cache)	Stores attention key-value pairs from previous tokens. Avoids recomputation — speeds up inference.
Prefix Cache	Caches the processed beginning of a prompt. Useful when system prompts are reused across many calls.
Prompt Caching	Saves processed prompt computation so repeated identical prompts skip full reprocessing.
Semantic Cache	Caches AI responses based on meaning similarity — semantically equivalent queries return cached answers.

MLOPS (23 terms)

Term	Description
Airflow (Apache)	Workflow orchestration for scheduling and monitoring data and ML pipelines.
AutoML	Automated model selection, hyperparameter tuning, pipeline optimization.
Canary Deployment	Rolling new model to small percentage of traffic first. Catches issues early.
Checkpoint (Training)	Saving model weights during training for resumption or rollback.
Concept Drift	Real-world data shifts after deployment, degrading model accuracy over time.
DVC (Data Version Control)	Git-like version control for datasets and ML models. Makes experiments reproducible.
Data Lineage	Tracking origin and all transformations of data through an ML pipeline.
Data Pipeline	Automated flow from data collection through cleaning, transformation, and training.

Data Versioning	Treating datasets like code — versioning with DVC for full reproducibility.
DataOps	Applying DevOps to data pipelines — automation, quality checks, version control for data.
Experiment Tracking	Logging hyperparameters, metrics, artifacts for every run. W&B, MLflow, Comet.
Feature Store	Centralized repository storing computed features for reuse across multiple models.
MLOps	DevOps applied to ML — automates training, evaluation, deployment, monitoring, retraining.
MLflow	Open-source ML lifecycle platform — experiment tracking, model registry, deployment.
Model Drift	Deployed model accuracy degrades as real-world data shifts from training distribution.
Model Registry	Central store for trained models with versioning and metadata. Like a package manager for AI.
Model Versioning	Tracking model iterations over time. Essential for reproducibility and rollback.
ModelOps	Operationalizing ML at enterprise scale — governance, lifecycle, compliance, risk management.
Pipeline Orchestration	Automating and scheduling ML workflows end to end.
Prefect	Python-native workflow orchestration. Simpler alternative to Airflow.
Shadow Mode	Running new model in parallel without affecting users — safe comparison before traffic switch.
Weights & Biases (W&B)	ML experiment tracking platform. Logs metrics, visualizes training, compares models.
ZenML	Open-source MLOps framework for portable, reproducible, production-ready ML pipelines.

LLMOPS (16 terms)

Term	Description
Arize AI	ML observability platform for monitoring model performance and data drift in production.
Cost per Token	Pricing metric for LLM API usage. Input and output tokens often priced differently.

Evaluation Pipeline	Automated quality checks on every model or prompt change before deployment. CI/CD for AI.
FinOps (AI)	Managing cloud AI costs — token budgets, model tier selection, caching strategies.
Guardrail Layer	Runtime safety checks — toxicity detection, PII filtering, topic restriction, policy enforcement.
Helicone	Open-source LLM observability — logs every request, tracks costs, latency, and errors.
LLM Gateway	Middleware routing LLM requests, enforcing rate limits, logging, caching, managing providers.
LLMOps	MLOps for LLMs — prompt versioning, eval pipelines, token cost tracking, model swapping.
LangFuse	Open-source LLM engineering platform for tracing, evaluations, and prompt management.
LangSmith	LangChain's observability platform — tracing LLM calls, testing prompts, debugging agents.
Observability (AI)	Monitoring AI in production — inputs, outputs, latency, token usage, drift, errors.
OpenTelemetry	Open standard for traces, metrics, logs from AI systems. Vendor-neutral observability.
Prompt Registry	Central repository storing, versioning, and managing prompts across teams.
Prompt Versioning	Tracking prompt changes over time like code — version control for system prompts.
Token Budget	Limits on tokens per AI call to control cost and enforce latency targets.
Tracing (AI)	Recording step-by-step execution — tool calls, retrievals, generations — for debugging.

AGENTICOPS (6 terms)

Term	Description
Agent Loop	The observe → think → act → observe cycle. Infinite loops with no progress are a key failure mode.
Agent Memory Management	Deciding what stays in context, what goes to external storage, what gets discarded.
AgenticOps	Operations discipline for agentic AI — orchestration, tool reliability, loop detection, cost control.

Human Escalation	Triggering handoff to human when agent confidence is low or task exceeds defined boundaries.
Loop Detection	Monitoring system detecting agent repeating actions without progress — prevents runaway costs.
Tool Reliability	Tracking success and failure rates of agent tool calls. Failed tools cause degradation and loops.

DEPLOYMENT (15 terms)

Term	Description
API (Application Programming Interface)	Protocols enabling software to communicate. Developers use APIs to integrate AI into products.
Batch Inference	Running predictions on large datasets all at once — efficient for non-realtime processing.
Downstream Task	Specific real-world application of a pretrained model — translation, classification, summarization.
Edge AI	AI on local devices — phones, sensors — for low-latency, private inference without cloud.
Federated Learning	Training AI across decentralized devices without sharing raw data. Preserves privacy.
GPU (Graphics Processing Unit)	Hardware enabling parallel computation. Essential for training and serving large AI models.
Inference	Running a trained model to generate predictions. The production phase after training.
Latency	Time between sending request and receiving response. Critical metric in production AI.
Model Serving	Infrastructure making models available via API for predictions. Torch Serve, Triton, BentoML.
Online Inference	Real-time predictions on individual requests as they arrive. Required for chat and APIs.
Quantization	Reducing model numerical precision (float32 to int4) to shrink size and speed up inference.
Speculative Decoding	Small draft model predicts tokens; large model verifies. Significant inference speedup.
TTFT (Time to First Token)	Delay before AI starts generating response. Critical for perceived responsiveness in chat.

Temperature	Controls output randomness. High = creative. Low = deterministic. Zero = always picks top token.
Throughput	Requests per second an AI system handles. Higher = lower cost per inference.

INFRASTRUCTURE (35 terms)

Term	Description
BentoML	Bundles ML model, preprocessing, and API for streamlined deployment as a service.
CI/CD Pipeline	Automates testing and safely deploying ML model updates to production.
Celery	Distributed task queue for Python. Manages async AI workloads and background inference jobs.
Docker	Containerization platform packaging AI apps with all dependencies for portable deployment.
Docker Compose	Defines and runs multi-container apps — model server + database + cache as one system.
Docker Container	Running instance of a Docker image. Isolated, lightweight, reproducible environment.
Docker Hub	Public registry for storing and sharing Docker images.
Docker Image	Immutable snapshot of container environment — OS, code, libraries. Built once, run anywhere.
Dockerfile	Script defining how to build a Docker image — base image, dependencies, entry point.
FastAPI	Python web framework for building high-performance AI APIs. Most popular for model serving.
GPU Operator	Kubernetes operator auto-configuring GPU drivers and plugins across cluster nodes.
Grafana	Visualization platform for monitoring dashboards displaying real-time AI system metrics.
Helm	Kubernetes package manager — Helm charts deploy complex AI applications on K8s.
Horizontal Pod Autoscaling (HPA)	Automatically scales K8s pods based on CPU, memory, or custom metrics like queue depth.

Infrastructure as Code (IaC)	Managing infrastructure through code (Terraform, Pulumi) for reproducibility.
Ingress Controller	Manages external HTTP/S access to K8s services. Routes requests to correct AI endpoints.
K8s Cluster	Set of nodes managed by Kubernetes. Control plane manages scheduling; workers run workloads.
K8s Deployment	Kubernetes object managing identical pod sets — automates rollouts, rollbacks, scaling.
K8s Node	Worker machine in a Kubernetes cluster. VM or physical server with CPU or GPU.
K8s Pod	Smallest deployable Kubernetes unit — one or more containers sharing network and storage.
K8s Service	Stable network endpoint exposing pods to traffic. Decouples app from pod IP addresses.
Kafka	Distributed event streaming platform feeding real-time data into ML pipelines.
Kubeflow	ML platform on Kubernetes for end-to-end ML pipelines — training, serving, experimentation.
Kubernetes (K8s)	Container orchestration platform automating deployment, scaling, management of AI workloads.
Load Balancer	Distributes incoming AI requests across model instances to prevent overload.
NVIDIA Triton	High-performance model serving for PyTorch, TensorFlow, ONNX. Optimized for GPU inference.
ONNX	Open Neural Network Exchange — standard format for sharing models across ML frameworks.
Prometheus	Open-source monitoring and alerting toolkit for tracking ML performance metrics.
REST API	Standard HTTP interface for exposing AI models. Simple, widely supported.
Ray	Distributed Python computing framework. Ray Serve deploys AI models at scale across clusters.
Redis	In-memory store used as semantic cache, session memory, message queue for AI apps.
Reverse Proxy	Sits in front of AI servers handling SSL, caching, routing, rate limiting. Nginx commonly used.

Terraform	HashiCorp's IaC tool for provisioning cloud AI infrastructure declaratively as code.
TorchServe	PyTorch's production model serving. Handles REST/gRPC APIs, batching, model management.
gRPC	High-performance RPC framework. Faster than REST for AI microservices. Used in Triton.

☁ AWS (21 terms)

Term	Description
AWS ECR	Elastic Container Registry — private Docker image registry on AWS for ML container images.
AWS EKS	Elastic Kubernetes Service — managed Kubernetes on AWS for containerized ML workloads.
AWS Glue	Serverless ETL for discovering, preparing, transforming data for ML training on AWS.
AWS Inferentia	AWS custom silicon for cost-efficient deep learning inference. Cheaper than GPU for serving.
AWS Lambda	Serverless compute for AI inference without managing servers. Pay per invocation.
AWS SageMaker	Amazon's fully managed ML platform — Studio, Autopilot, Pipelines, Endpoints in one place.
AWS Step Functions	Serverless workflow orchestrator for ML pipelines — coordinates services, handles retries.
AWS Trainium	AWS custom chip for training deep learning models. More cost-efficient than GPU instances.
Amazon Bedrock	Managed service for accessing foundation models (Claude, Llama, Titan) via API on AWS.
Amazon Comprehend	AWS NLP — sentiment analysis, entity recognition, topic modeling, language detection.
Amazon Kendra	AWS intelligent enterprise search powered by ML for natural language queries.
Amazon Lex	AWS service for building conversational chatbots. Core technology powering Alexa.
Amazon OpenSearch	AWS managed search and analytics for hybrid RAG pipelines combining keyword and vector search.

Amazon Polly	AWS text-to-speech service — natural-sounding audio in multiple languages.
Amazon Rekognition	AWS computer vision API — facial recognition, object detection, content moderation.
Amazon S3	Object storage for ML datasets, model artifacts, and outputs. Standard data lake on AWS.
Amazon Textract	Extracts text and structured data from scanned documents, forms, and tables.
Amazon Transcribe	AWS speech-to-text — real-time and batch transcription with speaker identification.
SageMaker Endpoints	Managed real-time inference endpoints. Auto-scaling, load balancing, health monitoring.
SageMaker Pipelines	MLOps service automating and orchestrating complete ML workflows on AWS.
SageMaker Studio	Web IDE for full ML lifecycle — notebooks, experiments, pipelines, deployment in one interface.

AZURE (20 terms)

Term	Description
Azure AI Content Safety	API detecting harmful content — violence, hate speech, sexual content — in AI inputs/outputs.
Azure AI Foundry	Microsoft's unified platform for building, deploying, managing enterprise AI and agent applications.
Azure AI Search	AI-powered enterprise search with vector search and semantic ranking. Primary RAG service on Azure.
Azure Blob Storage	Object storage for ML datasets and model artifacts on Azure. Equivalent to AWS S3.
Azure Bot Service	Platform for building, hosting, and connecting AI chatbots across Teams, web, and voice.
Azure Cognitive Services	Suite of pre-built AI APIs covering vision, speech, language, and decision — no ML expertise needed.
Azure Computer Vision	Pre-built image analysis — object detection, OCR, image captioning, spatial analysis.
Azure Container Registry (ACR)	Private Docker image registry on Azure. Integrates with AKS for ML deployments.
Azure Data Factory	Cloud ETL and data integration for building ML data ingestion and transformation pipelines.

Azure Databricks	Apache Spark analytics on Azure for large-scale ML training and data engineering.
Azure Functions	Serverless compute for AI inference and event-driven ML pipelines on Azure.
Azure Kubernetes Service (AKS)	Managed Kubernetes on Azure for deploying and scaling containerized ML workloads.
Azure Language Service	NLP API for sentiment, NER, summarization, and translation on Azure.
Azure ML Studio	Web UI for Azure ML — drag-and-drop pipeline builder and no-code ML interface.
Azure Machine Learning	Microsoft's end-to-end ML platform — experiments, pipelines, model registry, deployment, monitoring.
Azure OpenAI Service	Microsoft's managed API for OpenAI models (GPT-4, DALL-E, Whisper) within Azure's security boundary.
Azure Responsible AI Dashboard	Tool for model fairness, explainability, error analysis, causal inference on Azure.
Azure Speech Services	Speech-to-text, text-to-speech, and real-time speech translation APIs on Azure.
Azure Synapse Analytics	Unified analytics combining data warehousing and big data processing for ML on Azure.
Prompt Flow (Azure)	Azure tool for building, evaluating, deploying LLM-based flows. Integrated with Azure AI Foundry.

GCP (15 terms)

Term	Description
BigQuery ML	Run ML models directly in SQL inside Google BigQuery — no separate training infrastructure.
Gemini	Google's multimodal foundation model family. Available via Vertex AI and Google AI Studio.
Google AI Studio	Free browser-based tool for experimenting with Gemini models. Auto-generates API code.
Google Cloud Natural Language API	NLP for sentiment, entity, syntax, and classification on GCP.
Google Cloud Speech-to-Text	ASR supporting 125+ languages with custom model adaptation on GCP.

Google Cloud Storage (GCS)	Object storage for ML datasets and artifacts on GCP. Equivalent to S3 and Azure Blob.
Google Cloud TPU	Tensor Processing Units — Google's custom silicon for training and serving large ML models.
Google Cloud Translation	Neural machine translation API supporting 130+ languages.
Google Cloud Vision API	Pre-built image analysis — object labels, face detection, OCR, safe search, logo detection.
Google Dialogflow	Platform for building conversational AI — chatbots and voice agents — on GCP.
Google Kubernetes Engine (GKE)	Managed Kubernetes on GCP for large-scale ML serving and training workloads.
Google Vertex AI	Google's unified ML platform — training, serving, AutoML, monitoring, pipelines on GCP.
Vertex AI Model Garden	Catalog of foundation models on GCP — Gemini, open-source, and partner models.
Vertex AI Pipelines	Managed ML pipeline orchestration on GCP using Kubeflow Pipelines.
Vertex AI Workbench	Managed Jupyter notebook environment for ML development on Google Cloud.

PYTHON & TOOLS (24 terms)

Term	Description
Chroma	Open-source vector database for AI applications. Easy to embed locally or deploy in cloud.
FAISS	Facebook's library for efficient similarity search of dense vectors. Widely used in RAG.
Gradio	Library for building quick interactive ML demo UIs. Integrates with HuggingFace Spaces.
HuggingFace	Hub for open-source models, datasets, and demos. The GitHub of AI.
HuggingFace Transformers	Unified API for thousands of pretrained models. Most widely used ML library globally.
Jupyter Notebook	Browser-based interactive coding environment. Standard for ML experimentation.
LangChain	Framework for LLM-powered applications — chains, agents, RAG pipelines, tool integrations.

LiteLLM	Unified interface for 100+ LLM APIs with one consistent syntax. Lightweight LLM gateway.
LlamaIndex	Framework specializing in data ingestion, indexing, and querying for LLM and RAG.
Milvus	Open-source vector database for billion-scale similarity search. Cloud-native, high performance.
NumPy	Foundational Python numerical computing library. Underpins all scientific Python and ML.
Ollama	Tool for running open-source LLMs locally with a simple CLI. Supports Llama, Mistral, Phi.
Pandas	Python data manipulation library. Ubiquitous in data preprocessing for ML.
Pinecone	Managed cloud vector database. Fully hosted, production-grade semantic search infrastructure.
PyTorch	Meta's open-source ML framework. Dominant in research. Foundation of most modern AI models.
Python	Most used AI/ML programming language. Ecosystem: NumPy, Pandas, PyTorch, TensorFlow, HuggingFace.
Qdrant	High-performance vector database in Rust. Fast, scalable, open-source. Popular for production RAG.
R Programming Language	Statistical computing language popular in data science and academic ML.
Scikit-learn	Core Python ML library for classical algorithms — classification, regression, clustering.
Streamlit	Python framework for quickly building ML demos and data apps. No frontend knowledge needed.
TensorFlow	Google's ML framework. Dominant in production. TFLite for mobile and embedded AI.
Weaviate	Open-source vector database with built-in ML modules and hybrid search.
XGBoost	Extreme Gradient Boosting. High-performance ensemble library. Dominant for tabular data.
vLLM	High-throughput LLM inference using PagedAttention. Fastest open-source LLM serving framework.

DATA (15 terms)

Term	Description
Big Data	Datasets too large for traditional processing. Volume, velocity, variety — fuel for AI training.
Corpus	Large text collection for training language models. Quality, diversity, and size all matter.
Data Augmentation	Creating extra training examples by transforming existing data. Reduces overfitting.
Data Mining	Discovering patterns from large datasets using ML, statistics, and database methods.
Data Science	Extracting actionable insights from structured and unstructured data using statistics and ML.
Data Warehouse	Centralized repository integrating historical data from multiple sources for analytics and ML.
Dataset	Structured collection of data for training, validating, or testing AI models. Quality > quantity.
Feature	Measurable input variable used by ML models — word frequency, pixel value, sensor reading.
Feature Engineering	Manually creating or transforming features to improve ML model performance.
Metadata	Describes data — labels, timestamps, source. Stored with chunks in RAG for precise filtering.
Noise	Random or irrelevant data in training sets. Denoising separates signal from noise.
Test Data	Held-out data for final evaluation only. Never seen during training. True measure of generalization.
Training Data	Data teaching the model patterns. Garbage in = garbage out. Quality is the most critical factor.
Validation Data	Data for hyperparameter tuning during training. Separate from final test data.
Vector	List of numbers representing data mathematically. Semantic arithmetic operates on them.

COMPUTER VISION (5 terms)

Term	Description
Computer Vision	AI interpreting visual data from images and videos. Powers facial recognition, medical imaging.
Image Recognition	Identifying objects, people, or scenes in images. Foundation of self-driving, medical AI.
Machine Vision	Imaging-based AI for industrial inspection, quality control, and robot guidance.
Object Detection	Locating and classifying multiple objects in images with bounding boxes. Autonomous vehicles, robotics.
Vision-Language Model	AI understanding images and text together — describes images, reads charts, answers visual questions.

SAFETY (16 terms)

Term	Description
Adversarial Attack	Deliberately crafted inputs designed to fool or break an AI model — causes misclassification.
Alignment	Ensuring AI goals and actions match human values and intentions — the system does what we want.
Constitutional AI	Anthropic's training method guiding Claude with a set of principles toward safe, helpful, honest behavior.
Data Poisoning	Inserting malicious data into training sets to manipulate model behavior. Serious adversarial attack.
Guardrails	Safety rules preventing harmful AI outputs — applied at training time and at runtime.
Hallucination	AI confidently generating factually incorrect or fabricated information. Major reliability challenge.
Human-in-the-Loop	System requiring human review before AI decisions are acted on. Critical for high-stakes applications.
Interpretability	Research understanding AI model internals — what features activate, why specific outputs occur.
Jailbreak	Prompting attempting to bypass AI safety restrictions — tricking model into producing restricted content.
Memorization	Model reproducing training data verbatim. Privacy risk when sensitive data was used in training.

Prompt Injection	Attack embedding malicious instructions in content fed to AI — attempts to hijack behavior.
Prompt Leaking	Tricking AI into revealing its hidden system prompt — exposing proprietary instructions.
Red Teaming	Deliberately attacking AI to find safety vulnerabilities before public deployment.
Robustness	AI maintaining performance under adversarial attacks, distribution shift, or noisy inputs.
Safety (AI)	Ensuring AI doesn't cause harm to individuals, society, or the world by accident or misuse.
Sandboxing	Running AI-generated code in isolated environment to prevent affecting real production systems.

ETHICS (10 terms)

Term	Description
AI Forensics	Identifying whether text is AI-generated, tracing its source, revealing bias in training data.
AI Governance	Frameworks ensuring ethical, accountable, compliant AI development and deployment.
AI Literacy	Competency to understand, evaluate, use, and critically assess AI. Essential in every profession.
Bias (AI / Algorithmic)	Systematic unfairness from skewed training data causing AI to discriminate against groups.
Data Privacy	Protecting sensitive information in AI pipelines. Differential privacy, anonymization techniques.
Explainability	Ability to explain why AI made a specific decision. Required for regulated domains.
Explainable AI (XAI)	Techniques making black-box models interpretable — SHAP, LIME, attention visualization.
Fairness	Ensuring AI treats all demographic groups equitably. Demographic parity, equal opportunity metrics.
Model Card	Document describing capabilities, training data, intended uses, limitations, and known risks.
Responsible AI	Developing AI with fairness, transparency, privacy, human oversight as core design principles.

EVAL (12 terms)

Term	Description
A/B Testing (AI)	Running two model versions with real users to compare performance metrics.
Ablation Study	Removing a component from AI to measure how much it contributes to overall performance.
BLEU Score	Evaluates machine-generated text by measuring n-gram overlap with human references.
Benchmark	Standardized test measuring AI performance. MMLU, HumanEval, HELM, BIG-bench, GLUE.
Ecological Validity	Degree to which controlled evaluation results transfer to real-world deployment conditions.
Evaluation (Eval)	Systematically testing model performance to determine production readiness.
Faithfulness	How accurately AI answer reflects source material without distortion or fabrication.
Groundedness	Whether AI response is supported by source documents. Primary RAG quality metric.
Needle in a Haystack	Benchmark hiding a fact deep in long document to test true long-context retrieval ability.
Perplexity	Measures LLM confidence in text prediction. Lower = more certain. Common intrinsic metric.
RAGAS	Framework evaluating RAG pipelines — faithfulness, answer relevance, context recall, precision.
Relevance Score	Measures how relevant retrieved documents are to user query. Key RAG evaluation metric.

REASONING (32 terms)

Term	Description
Abductive Reasoning	Starts from observations, finds simplest likely explanation. Yields plausible but not certain conclusions.
Automated Planning	Finding action sequences to achieve goals. Used in robotics, logistics, game AI.
Backward Chaining	Works backward from goal to find supporting facts. Used in expert systems and logic programming.

Bayesian Programming	Specifying probabilistic models for problems with incomplete information. Represents uncertainty explicitly.
Blackboard System	Multiple specialist modules collaborating via a shared 'blackboard' knowledge base.
Case-Based Reasoning	Solving new problems by finding similar past problems and adapting their solutions.
Cognitive Architecture	Blueprint for intelligent systems describing fixed structures giving rise to intelligent behavior.
Cognitive Science	Interdisciplinary study of the mind bridging AI, psychology, linguistics, and neuroscience.
Commonsense Knowledge	Facts about everyday world all humans know. Fundamental for AI interacting naturally with people.
Commonsense Reasoning	AI making intuitive assumptions about everyday situations that aren't explicitly stated.
Decision Support System (DSS)	Helps humans make decisions about complex problems combining data, models, and interfaces.
First-Order Logic	Formal logic using quantified variables and predicates. Foundation of knowledge representation.
Forward / Backward Chaining	Forward derives conclusions from facts; backward starts from goals and finds supporting facts.
Frame Problem	Challenge of representing what stays unchanged when actions occur. Fundamental in AI planning.
Fuzzy Logic	Logic handling degrees of truth between 0 and 1. Used in control systems and uncertain decisions.
Inference Engine	Software applying logical rules to a knowledge base to derive new information.
Knowledge Base	Structured repository of facts, rules, and relationships used by AI for reasoning or retrieval.
Knowledge Engineering	Building knowledge-based systems — capturing expert knowledge into formal representations.
Knowledge Graph	Graph-structured database of entities and relationships. Used in GraphRAG, semantic search.
Knowledge Representation	Formal encoding of world knowledge enabling AI reasoning — ontologies, logic, semantic networks.
Markov Chain	Stochastic model where each state depends only on previous state. Foundation of probabilistic AI.

NP-Completeness	A problem class relevant to understanding computational limits of AI planning and search.
Occam's Razor	When models perform equally, prefer the simpler one. Guides regularization and model selection.
Ontology Learning	Automatic creation of ontologies from text — extracting domain concepts and relationships.
Partial Order Reduction	Reduces state-space in planning by exploiting commutativity of concurrent independent actions.
Probabilistic Programming	Programming paradigm where probabilistic models are specified and inference runs automatically.
Production System	AI using condition-action rules to match patterns and trigger responses. Foundation of expert systems.
Rule-Based System	AI using human-crafted if-then rules. Interpretable but brittle and hard to scale.
Semantic Network	Knowledge representation using concept nodes and relationship edges. Predecessor to knowledge graphs.
Spatial-Temporal Reasoning	AI reasoning about space and time simultaneously. Used in robotics and autonomous navigation.
Symbolic AI	AI using human-readable symbolic representations — logic, rules, knowledge graphs — not numeric weights.
Turing Machine	Abstract model of computation defining theoretical limits of what computers can compute.

OPTIMIZATION (12 terms)

Term	Description
Ant Colony Optimization (ACO)	Probabilistic optimization inspired by ant foraging behavior. Finds good paths through graphs.
Evolutionary Algorithm	Optimization inspired by biological evolution — selection, mutation, crossover over generations.
Genetic Algorithm (GA)	Metaheuristic using selection, crossover, and mutation to optimize solutions — inspired by evolution.
Heuristic	Problem-solving technique trading optimality for speed — finds good-enough solutions fast.
Metaheuristic	High-level optimization strategy for finding solutions when exhaustive search is infeasible.

Monte Carlo Tree Search	Heuristic search using random sampling through decision trees. Used in game AI and planning.
Particle Swarm Optimization	Particles move through solution space guided by personal and global best positions.
Search Algorithm	Systematically explores a problem space to find solutions. Foundation of game AI and planning.
Simulated Annealing	Occasionally accepts worse solutions to escape local optima — inspired by physical annealing.
Stochastic Optimization	Optimization using random variables. Foundation of SGD — the primary deep learning training algorithm.
Swarm Intelligence	Collective behavior of decentralized agents inspiring optimization. Ant colonies, particle swarms.
Thompson Sampling	Heuristic for exploration-exploitation tradeoff — samples from posterior beliefs to choose actions.

ROBOTICS (6 terms)

Term	Description
Autonomous Robot	Robot performing tasks with high autonomy — perceiving, deciding, acting without constant human control.
Autonomous Vehicle	Vehicle capable of sensing environment and navigating with little or no human input. Self-driving car.
Cloud Robotics	Robots leveraging cloud AI — offloading heavy computation to cloud for smarter, lighter physical systems.
Neurocybernetics / BCI	Brain-computer interface providing direct communication between brain and external devices.
Robotics	AI combined with physical machines to perceive, decide, and act in the real world.
Sensor Fusion	Combining data from multiple sensors for more accurate information than any single sensor provides.

COMPANY (10 terms)

Term	Description
AlphaGo	Google DeepMind's program that beat world champion Go player in 2016. First AI to defeat a professional without handicap.

Anthropic	AI safety company that created Claude. Focused on building safe, beneficial, understandable AI.
ChatGPT	OpenAI's LLM chatbot launched November 2022. Brought large language models to mainstream awareness.
Claude	Anthropic's AI assistant. Designed to be helpful, harmless, and honest. Trained with Constitutional AI and RLHF.
Dartmouth Workshop	The 1956 summer workshop widely considered the founding event of artificial intelligence as a field.
Deep Blue	IBM's chess computer defeating world champion Garry Kasparov in 1997. First AI to beat a reigning world chess champion.
DeepMind	Alphabet/Google AI lab. Created AlphaGo, AlphaFold, and Gemini. Pioneered deep reinforcement learning.
GPT (Generative Pre-trained Transformer)	OpenAI's flagship LLM family. Pre-trained on internet text then fine-tuned with RLHF. Powers ChatGPT.
OpenAI	AI company behind GPT, ChatGPT, DALL-E, Whisper. Originally nonprofit, now major commercial AI lab.
Watson	IBM's AI — won Jeopardy! in 2011. Pioneer of enterprise AI and question-answering at scale.

FINAL MASTER COUNT

Category	Count
Core AI	22
Concept	19
Machine Learning	42
Deep Learning	14
Architecture	22
NLP	17
GenAI	5
RAG	16
Prompting	14
Agents	13
Caching	6

MLOps	23
LLMOps	16
AgenticOps	6
Deployment	15
Infrastructure	35
AWS	21
Azure	20
GCP	15
Python & Tools	24
Data	15
Computer Vision	5
Safety	16
Ethics	10
Eval	12
Reasoning	32
Optimization	12
Robotics	6
Company	10
Training	20
TOTAL	**502**